

datsi

DV 2026/2027

Data Visualization: Visualization Project

Pablo Toharia

Contents

1 Introduction **2**
- 1.1 Getting support . . . 2
- 1.2 Delivering the project . . . 2
- 1.3 Deadlines . . . 2
- 1.4 Usage of AI tools . . . 2

2 Design of a new interactive data analysis tool **3**
- 2.1 Design and develop steps . . . 3
- 2.2 Programming language and environment . . . 4
- 2.3 Minimum requirements and extra possible work . . . 4
- 2.4 Report . . . 5

References **6**

1 Introduction

The objective of this project is to help students to put into practice the concepts learnt during the theory classes and to get proficiency in the use of some tools that will allow them to interactively analyze a set of data. Furthermore, students will be able to develop their own tools to conduct this kind of analysis.

1.1 Getting support

You can contact the lecturer in charge of the practical project at the following email address:

- Pablo Toharia <pablo.toharia@upm.es>

1.2 Delivering the project

The project will be sent through the corresponding submission form in Moodle. All the files must be compressed in a single `.zip` file before submitting them. The project must be carried out in groups of four or five people and only one of the members of the team needs to submit the practical project.

You do not need to provide any printed document. The project will be evaluated electronically. Apart from the source code, the delivery must have a PDF file with a report of the work done including: data abstraction, task abstraction, idioms used, justification of the decisions made, etc.

> DO NOT NEGLECT THE QUALITY OF THE REPORT: It will be evaluated in the same way the source code will be and needs to obtain a favorable evaluation in order for the practical project to pass. A low quality report may make you fail the practical project, even if the source code works properly.

1.3 Deadlines

The deadlines are the following:

- Approval of the chosen dataset: Oct 5th, 2026.
- Project delivery: Jan 11th, 2027.

1.4 Usage of AI tools

The use of AI-based tools is permitted during the development of the practical project. However, students are expected to understand, justify, and critically discuss all methodological, analytical, and design decisions reflected in their work. The project report must

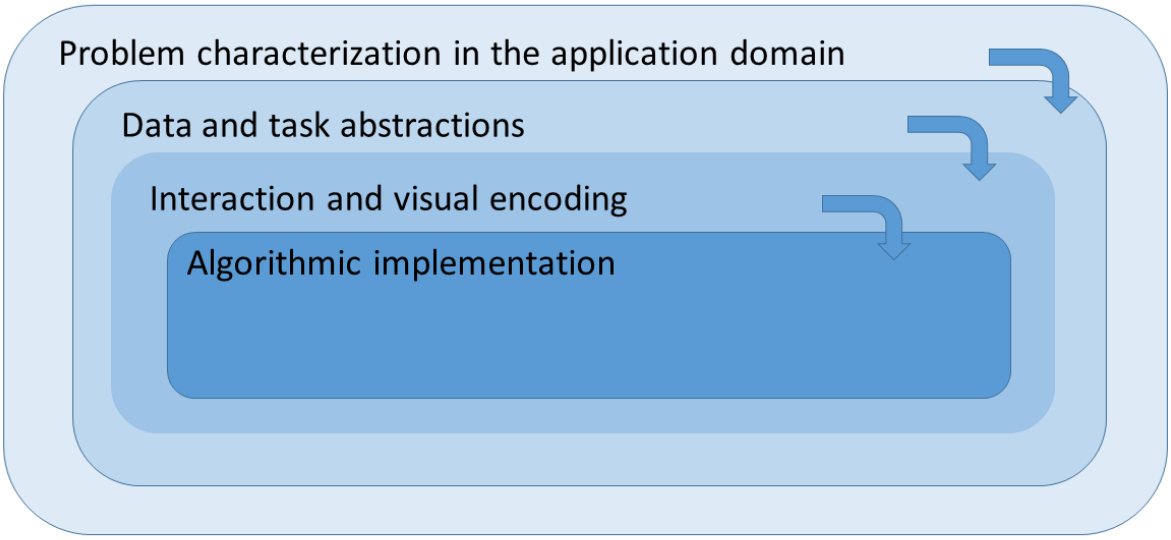


Figure 1: Design abstraction levels

include a declaration specifying which AI-based tools were used, how they were used, and which parts of the work were assisted by such tools.

The final written exam may include questions related to the practical project, and students should be able to explain and defend their project.

2 Design of a new interactive data analysis tool

This exercise will consist in the design and implementation of an interactive visualization tool. The tool should address the specific analytical problems that an end user or analyst faces when exploring and analyzing a given dataset. The analysis and design should follow the methodological steps presented in Figure 1 and discussed in the theory classes.

2.1 Design and develop steps

The first step is to characterize the problem in the application domain. For that, students should select a dataset and pose some questions the analysts would like to answer with the visual analytics tool. In order to make the level of complexity uniform for all the students, the selected dataset should be approved (in class time or by email/Teams) by the lecturer responsible of the practical work (check deadline in Section 1.3).

Once the selection of the dataset has been validated by the lecturer, the students must pose the questions (tasks) the potential user of the tool (the "analyst") could ask. The questions should be well suited to being answered through visualization. Avoid questions that could be answered more effectively by other means, such as querying a table.

Once the data and tasks are defined in the application domain, the next step is to do the data abstraction and the task abstraction (as seen in theory classes).

When the data and task abstractions have been defined, students should start the design of the tool by choosing idioms (visualization+interaction) that are suited and effective for the tasks and data.

Finally, the design has to be implemented and published.

2.2 Programming language and environment

The recommended setup for the project is R as the programming language, RStudio as the development environment, and Shiny as the application framework. Alternatively, Python can be used as the main programming language, using Shiny for Python. Another option is to use JavaScript, D3 and Observable.

2.3 Minimum requirements and extra possible work

The questions and design must comply, at least, with the following **minimum** constraints:

- The tool should use at least 3 different visualization idioms that effectively support the considered questions (4 idioms if the group is formed by 5 people).
- The visualizations should provide appropriate interaction mechanisms that support the analysts' tasks. Each visualization idiom should incorporate at least one meaningful interaction mechanism (e.g., filtering, selection, aggregation, navigation, zooming, etc.).
- The application must be published online using one of the following platforms: Shinyapps.io or Posit Connect Cloud for Shiny applications, or Observable for JavaScript/D3 applications. Note: Links to the applications may be published on the instructor's or course's web page. If you do not want your application to be listed, please contact the instructor.

These are the minimum requirements to pass the practical work. Students may obtain a higher evaluation by incorporating additional design features, particularly those discussed in class, when they are properly justified by the analytical tasks. Examples include:

- Use of multiple (synchronized) views.
- Integration of the application with some data analysis algorithm (clustering analysis, for example).
- Use of specific arrangements for the data under consideration (mainly geographic arrangement for the datasets with that information, but only if it is justified by the task!).

2.4 Report

The delivered report should meet the following requirements:

- Structure the report according to the four design abstraction levels (Figure 1), and all the decisions made at every level must be discussed, highlighting those dealing with the **interaction level and the visual encoding**. To this purpose, students should be aware that the data and tasks that the analyst wants to carry out are the base on top of which these decisions must be justified.
- Include clear instructions on how to access and use the tool, together with any information or dependencies required for its evaluation. The report should also provide instructions for running the application locally, if applicable.
- Include the URL of the published application in the report.
- Include screenshots of the different idioms that illustrate how they support the questions posed by the analysts.

Revision History

This document has been published as a teaching report by the *Departamento de Arquitectura y Tecnología de Sistemas Informáticos* of the *Escuela Técnica Superior de Ingenieros Informáticos*, *Universidad Politécnica de Madrid* as part of the course *Information Visualization*. The history of changes made to this document is:

Revision	Date	Author(s)	Description
1.1	09/09/2015	JM, AR	Original version
1.2	30/09/2015	AR	New latex
2.1	30/11/2015	ALT, JM	New contents
2.2	17/11/2016	ALT	Updated statement
3.1	6/2/2019	PT	Changed structure
3.2	6/11/2019	PT	Dates and subject names changed
3.3	14/11/2020	PT	Updated dates and other minor changes
3.4	11/11/2021	PT	Updated dates and rewritten part of the second exercise.
3.5	14/10/2022	PT	Updated dates tune minor aspects about the second exercise.
3.6	11/09/2023	PT	Updated dates, logo, and removed BD references.
3.7	19/09/2023	PT	Added the possibility of groups of 4.
3.8	9/09/2024	PT	Updated dates. Fixed typos.
3.9	4/09/2025	PT	Updated dates.

4.0	7/09/2026	PT	Simplified document and adapted to new learning guide.

This document is currently maintained by: Pablo Toharia. © Departamento de Arquitectura y Tecnología de Sistemas Informáticos, Escuela Técnica Superior de Ingenieros Informáticos

References

[1] R Core Team: *The R Project for Statistical Computing*. https://www.r-project.org/

[2] Posit: *RStudio*. https://posit.co/products/open-source/rstudio/

[3] Posit: *Shiny*. https://shiny.posit.co/

[4] Posit: *Shiny for Python*. https://shiny.posit.co/py/

[5] Posit: *shinyapps.io*. https://www.shinyapps.io/

[6] Posit: *Posit Connect Cloud*. https://connect.posit.cloud/

[7] Bostock, M., Ogievetsky, V., Heer, J.: *D3: Data-Driven Documents*. https://d3js.org/

[8] Observable: *Observable*. https://observablehq.com/